

Namespace TUIS

Classes

[DynamicTerminal](#)

[Terminal](#)

A Terminal is the main object of this TUI system. It will hold each [Component](#) that will be drawn on the screen.

Enums

[Terminal.BackgroundColor](#)

Get the [Terminal.BackgroundColor](#) using the ANSI color code.

[Terminal.TextColor](#)

Get the [Terminal.TextColor](#) using th ANSI color code.

[Terminal.TextDecoration](#)

Get the [Terminal.TextDecoration](#) using the ANSI color code.

Class DynamicTerminal

Namespace: [TUIS](#)

Assembly: TUIS.dll

```
public class DynamicTerminal : Terminal
```

Inheritance

[object](#)  ← [Terminal](#) ← DynamicTerminal

Inherited Members

[Terminal.Width](#) , [Terminal.Height](#) , [Terminal.InputSystem](#) , [Terminal.TimeSystem](#) ,
[Terminal.Components](#) , [Terminal.NeedReDraw](#) ,
[Terminal.DrawChar\(int, int, char, Terminal.TextColor, Terminal.BackgroundColor, Terminal.Text
Decoration\)](#) ,
[Terminal.DrawChar\(int, int, char, \(int r, int g, int b\), Terminal.BackgroundColor, Terminal.TextDecoration\)](#) ,
[Terminal.Clear\(\)](#) , [Terminal.Start\(Action<Terminal>\)](#) , [object.Equals\(object\)](#)  ,
[object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

DynamicTerminal(int, int, ConsoleKeyInfo)

```
public DynamicTerminal(int width, int height, ConsoleKeyInfo switchKeyInfo)
```

Parameters

width [int](#) 

height [int](#) 

switchKeyInfo [ConsoleKeyInfo](#) 

Fields

SelectableComponents

```
public List<SelectableComponent> SelectableComponents
```

Field Value

[List](#) [<SelectableComponent>](#)

Methods

AddComponent(Component)

```
public override void AddComponent(Component component)
```

Parameters

component [Component](#)

Class Terminal

Namespace: [TUIS](#)

Assembly: TUIS.dll

A Terminal is the main object of this TUI system. It will hold each [Component](#) that will be drawn on the screen.

```
public class Terminal
```

Inheritance

[object](#)  ← Terminal

Derived

[DynamicTerminal](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Terminal(int, int)

The terminal is the class that holds every [Components](#) together, it also holds the [InputSystem](#) and the [TimeSystem](#). You should only have 1 instance of a terminal in your program.

```
public Terminal(int width = -1, int height = -1)
```

Parameters

width [int](#) 

Width of the terminal (default = -1 = [WindowWidth](#) ) (!\ cannot be changed).

height [int](#) 

Height of the terminal (default = -1 = [WindowHeight](#) ) (!\ cannot be changed).

Fields

Components

```
public readonly List<Component> Components
```

Field Value

[List](#) [<Component>](#)

Height

```
public readonly int Height
```

Field Value

[int](#)

InputSystem

```
public readonly InputSystem InputSystem
```

Field Value

[InputSystem](#)

NeedReDraw

```
public bool NeedReDraw
```

Field Value

[bool](#)

TimeSystem

```
public readonly TimeSystem TimeSystem
```

Field Value

[TimeSystem](#)

Width

```
public readonly int Width
```

Field Value

[int](#)

Methods

AddComponent(Component)

```
public virtual void AddComponent(Component component)
```

Parameters

component [Component](#)

Clear()

```
public void Clear()
```

DrawChar(int, int, char, (int r, int g, int b), BackgroundColor, TextDecoration)

```
public void DrawChar(int y, int x, char c, (int r, int g, int b) textColor,  
Terminal.BackgroundColor backgroundColor, Terminal.TextDecoration textDecoration)
```

Parameters

y [int](#)

x [int](#)

c [char](#)

textColor ([int](#) r, [int](#) g, [int](#) b)

backgroundColor [Terminal.BackgroundColor](#)

textDecoration [Terminal.TextDecoration](#)

DrawChar(int, int, char, TextColor, BackgroundColor, TextDecoration)

This method will be called by [DrawChar\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#) to correctly update a character of the internal TUIS.Terminal._screen of the [Terminal](#).

```
public void DrawChar(int y, int x, char c, Terminal.TextColor textColor,  
Terminal.BackgroundColor backgroundColor, Terminal.TextDecoration textDecoration)
```

Parameters

y [int](#)

The y coordinate of the char to update (0 = top).

x [int](#)

The x coordinate of the char to update (0 = left).

c [char](#)

The new char to draw.

textColor [Terminal.TextColor](#)

The [Terminal.TextColor](#) the **c** will be drawn.

backgroundColor [Terminal.BackgroundColor](#)

The [Terminal.BackgroundColor](#) the **c** will be drawn.

textDecoration [Terminal.TextDecoration](#)

The [Terminal.TextDecoration](#) the **c** will be drawn.

Start(Action<Terminal>?)

```
public void Start(Action<Terminal>? onStart = null)
```

Parameters

onStart [Action](#) <[Terminal](#)>

Enum Terminal.BackgroundColor

Namespace: [TUIS](#)

Assembly: TUIS.dll

Get the [Terminal.BackgroundColor](#) using the ANSI color code.

```
public enum Terminal.BackgroundColor
```

Fields

Black = 40

Blue = 44

Cyan = 46

Green = 42

None = 0

Purple = 45

Red = 41

White = 47

Yellow = 43

Enum Terminal.TextColor

Namespace: [TUIS](#)

Assembly: TUIS.dll

Get the [Terminal.TextColor](#) using th ANSI color code.

```
public enum Terminal.TextColor
```

Fields

Black = 30

Blue = 34

BrightBlue = 94

BrightCyan = 96

BrightGreen = 92

BrightPurple = 95

BrightRed = 91

BrightWhite = 97

BrightYellow = 93

Cyan = 36

Gray = 90

Green = 32

Purple = 35

Red = 31

White = 37

Yellow = 33

Enum Terminal.TextDecoration

Namespace: [TUIS](#)

Assembly: TUIS.dll

Get the [Terminal.TextDecoration](#) using the ANSI color code.

```
public enum Terminal.TextDecoration
```

Fields

Bold = 1

Default = 0

Underline = 4

Namespace TUIS.Components

Classes

[Component](#)

[SelectableComponent](#)

Class Component

Namespace: [TUIS.Components](#)

Assembly: TUIS.dll

```
public class Component
```

Inheritance

[object](#)  ← Component

Derived

[SelectableComponent](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Component(Terminal, int, int, int, int)

A Component will represent an element drawn on the [Terminal](#).

```
public Component(Terminal terminal, int width, int height, int posY, int posX)
```

Parameters

terminal [Terminal](#)

The [Terminal](#) the Component will be attached to.

width [int](#) 

The width of the component (-1 = [Width](#)).

height [int](#) 

The height of the component (-1 = [Height](#)).

posY [int](#)

The position on the y axis of the top left most char of the component (0 = top, -1 = will center the component).

posX [int](#)

The position on the x axis of the top left most char of the component (0 = left, -1 = will center the component).

Fields

Masks

```
public readonly List<Mask> Masks
```

Field Value

[List](#) <[Mask](#)>

Terminal

```
public Terminal Terminal
```

Field Value

[Terminal](#)

Properties

Height

```
public int Height { get; set; }
```

Property Value

[int](#)

IsVisible

```
public bool IsVisible { get; set; }
```

Property Value

[bool](#)

NeedReDraw

```
public bool NeedReDraw { get; set; }
```

Property Value

[bool](#)

PosX

```
public int PosX { get; set; }
```

Property Value

[int](#)

PosY

```
public int PosY { get; set; }
```

Property Value

[int](#)

Width

```
public int Width { get; set; }
```

Property Value

[int](#)

Methods

Draw()

```
public void Draw()
```

Class SelectableComponent

Namespace: [TUIS.Components](#)

Assembly: TUIS.dll

```
public class SelectableComponent : Component
```

Inheritance

[object](#)  ← [Component](#) ← SelectableComponent

Inherited Members

[Component.Terminal](#) , [Component.Width](#) , [Component.Height](#) , [Component.PosX](#) , [Component.PosY](#) , [Component.Masks](#) , [Component.NeedReDraw](#) , [Component.IsVisible](#) , [Component.Draw\(\)](#) , [object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

SelectableComponent(Terminal, int, int, int, int)

```
public SelectableComponent(Terminal terminal, int width, int height, int posY, int posX)
```

Parameters

terminal [Terminal](#)

width [int](#) 

height [int](#) 

posY [int](#) 

posX [int](#) 

Properties

IsSelected

```
public virtual bool IsSelected { get; set; }
```

Property Value

[bool](#)

Namespace TUIS.Components.Masks

Classes

[BackgroundMask](#)

Print a single char in the background (Warning Masks will be printed in order so this one will override any Mask before itself)

[BigTextMask](#)

This [Mask](#) will draw literal custom text using ascii art.

[BoxMask](#)

This [Mask](#) will draw a box on the border of the [Component](#).

[ClockMask](#)

This [Mask](#) will draw a clock and updates it accordingly to the real time.

[CustomMask](#)

This [Mask](#) helps to create a [Mask](#) with unique behaviour without needing to create an entire new class.

[ImageMask](#)

This Mask will draw an image in format jpg, png or any other image format in ASCII art in black and white or using 8 different colors.

[InputMask](#)

WIP. This mask let you use [ReadLine\(\)](#)  to get some user input as command or text

[Mask](#)

A Mask must be attached to a [Component](#) and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the [Masks](#) list.

[TextMask](#)

This [Mask](#) will draw literal custom text.

Enums

[BigTextMask.FontType](#)

Enum representing the different font types possible.

[BoxMask.Type](#)

The Type of boxing

[ImageMask.ImageCharacterisation](#)

[TextMask.HorizontalAlignmentEnum](#)

[TextMask.VerticalAlignmentEnum](#)

Class BackgroundMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

Print a single char in the background (Warning Masks will be printed in order so this one will override any Mask before itself)

```
public class BackgroundMask : Mask
```

Inheritance

[object](#)  ← [Mask](#) ← BackgroundMask

Inherited Members

[Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) ,
[Mask.Decoration](#) , [Mask.Draw\(\)](#) ,
[Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

BackgroundMask(Component, char, bool, TextColor,
BackgroundColor, TextDecoration)

```
public BackgroundMask(Component component, char backgroundChar = ' ', bool isVisible  
= true, Terminal.TextColor color = TextColor.White, Terminal.BackgroundColor  
background = BackgroundColor.None, Terminal.TextDecoration decoration  
= TextDecoration.Default)
```

Parameters

component [Component](#)

backgroundChar [char](#) 

isVisible [bool](#)

color [Terminal.TextColor](#)

background [Terminal.BackgroundColor](#)

decoration [Terminal.TextDecoration](#)

Properties

BackgroundChar

Holds the character to be drawn as the background.

```
public char BackgroundChar { get; set; }
```

Property Value

[char](#)

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

This [Mask](#) will draw the [BackgroundChar](#) on top of the intire [Component](#).

```
protected override void Behaviour()
```

Class BigTextMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

This [Mask](#) will draw literal custom text using ascii art.

```
public class BigTextMask : TextMask
```

Inheritance

[object](#)  ← [Mask](#) ← [TextMask](#) ← BigTextMask

Inherited Members

[TextMask.Text](#) , [TextMask.HorizontalPadding](#) , [TextMask.VerticalPadding](#) , [TextMask.VerticalAlignment](#) , [TextMask.HorizontalAlignment](#) , [Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) , [Mask.Decoration](#) , [Mask.Draw\(\)](#) , [Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.TextDecoration?\)](#) , [Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.TextDecoration?\)](#) , [object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

BigTextMask(Component, string, FontType, byte, byte, HorizontalAlignmentEnum, VerticalAlignmentEnum, bool, TextColor, BackgroundColor, TextDecoration)

This [TextMask](#) will draw the text using ascii art.

```
public BigTextMask(Component component, string text, BigTextMask.FontType fontType = FontType.ThreeByThreeClassic, byte horizontalPadding = 0, byte verticalPadding = 0, TextMask.HorizontalAlignmentEnum horizontalAlignment = HorizontalAlignmentEnum.Left, TextMask.VerticalAlignmentEnum verticalAlignment = VerticalAlignmentEnum.Top, bool isVisible = true, Terminal.TextColor color = TextColor.White, Terminal.BackgroundColor background = BackgroundColor.None, Terminal.TextDecoration decoration = TextDecoration.Default)
```

Parameters

component [Component](#)

The component which the mask is attached to.

text [string](#)

The string of text to draw.

fontType [BigTextMask.FontType](#)

The type of font to draw.

horizontalPadding [byte](#)

The distance between the left/right sides of the **component** and the beginning/end of the **text**'s rows.

verticalPadding [byte](#)

The distance between the top/bottom sides of the **component** and the first/last row of the **text**.

horizontalAlignment [TextMask.HorizontalAlignmentEnum](#)

Whether the **text** should be aligned to the HorizontalAlignmentEnum.Left, HorizontalAlignmentEnum.Center or HorizontalAlignmentEnum.Right of the **component**.

verticalAlignment [TextMask.VerticalAlignmentEnum](#)

Whether the **text** should be aligned to the VerticalAlignmentEnum.Top, VerticalAlignmentEnum.Center or VerticalAlignmentEnum.Bottom of the **component**.

isVisible [bool](#)

Represent the visibility of the mask (default = true).

color [Terminal.TextColor](#)

The default color of the mask (a mask's [Behaviour\(\)](#) method can override the color (default = white).

background [Terminal.BackgroundColor](#)

The default background color of the mask (a mask's [Behaviour\(\)](#) method can override the background color (default = None).

decoration [Terminal.TextDecoration](#)

The default decoration of the mask (a mask's [Behaviour\(\)](#) method can override the decoration (default = Default).

Properties

Font

Holds the current [BigTextMask.FontType](#).

```
public BigTextMask.FontType Font { get; set; }
```

Property Value

[BigTextMask.FontType](#)

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

```
protected override void Behaviour()
```

Enum BigTextMask.FontType

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

Enum representing the different font types possible.

```
public enum BigTextMask.FontType
```

Fields

```
ThreeByThreeClassic = 0
```

Class BoxMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

This [Mask](#) will draw a box on the border of the [Component](#).

```
public class BoxMask : Mask
```

Inheritance

[object](#)  ← [Mask](#) ← BoxMask

Inherited Members

[Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) ,
[Mask.Decoration](#) , [Mask.Draw\(\)](#) ,
[Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[object.Equals\(object\) !\[\]\(4754fc919b2e8116c30595fd4b918f00_img.jpg\)](#) , [object.Equals\(object, object\) !\[\]\(398309e51ba3618e371a2962765ad1a5_img.jpg\)](#) , [object.GetHashCode\(\) !\[\]\(afa25053f537a25280bb6941463adad1_img.jpg\)](#) , [object.GetType\(\) !\[\]\(07329cd37651c8207340e25e3ac65499_img.jpg\)](#) ,
[object.MemberwiseClone\(\) !\[\]\(33928686c0fa41d62d6b2537e1fbe49a_img.jpg\)](#) , [object.ReferenceEquals\(object, object\) !\[\]\(de560eefac7f98cac84b7d6fb97b06e9_img.jpg\)](#) , [object.ToString\(\) !\[\]\(dafd27a0ba4ce99d00ab8eb13d49bf62_img.jpg\)](#)

Constructors

BoxMask(Component, Type, bool, TextColor, BackgroundColor, TextDecoration)

Add a bounding box on the component.

```
public BoxMask(Component component, BoxMask.Type type, bool isVisible = true,  
Terminal.TextColor color = TextColor.White, Terminal.BackgroundColor background =  
BackgroundColor.None, Terminal.TextDecoration decoration = TextDecoration.Default)
```

Parameters

component [Component](#)

type [BoxMask.Type](#)

The [BoxMask.Type](#) of boxing.

isVisible [bool](#)

color [Terminal.TextColor](#)

background [Terminal.BackgroundColor](#)

decoration [Terminal.TextDecoration](#)

Properties

BoxType

Holds the [BoxMask.Type](#) of boxing.

```
public BoxMask.Type BoxType { get; set; }
```

Property Value

[BoxMask.Type](#)

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

This [Mask](#) will draw a box on the border of the [Component](#).

```
protected override void Behaviour()
```

Enum BoxMask.Type

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

The Type of boxing

```
public enum BoxMask.Type
```

Fields

Bold = 2

Default = 0

Double = 3

None = 6

Rounded = 1

Thick = 4

ThickInner = 5

Class ClockMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

This [Mask](#) will draw a clock and updates it accordingly to the real time.

```
public class ClockMask : Mask
```

Inheritance

[object](#)  ← [Mask](#) ← ClockMask

Inherited Members

[Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) ,
[Mask.Decoration](#) , [Mask.Draw\(\)](#) ,
[Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

ClockMask(Component, bool, TextColor, BackgroundColor, TextDecoration)

Add a real time clock to the [Component](#). Warning: The clock is 27 wide by 3 tall. Any [Component](#) smaller than that will cause the clock to overshoot.

```
public ClockMask(Component component, bool isVisible = true, Terminal.TextColor  
color = TextColor.White, Terminal.BackgroundColor background = BackgroundColor.None,  
Terminal.TextDecoration decoration = TextDecoration.Default)
```

Parameters

component [Component](#)

The component which the mask is attached to.

`isVisible` [bool](#)

Represent the visibility of the mask (default = true).

`color` [Terminal.TextColor](#)

The default color of the mask (a mask's [Behaviour\(\)](#) method can override the color (default = white).

`background` [Terminal.BackgroundColor](#)

The default background color of the mask (a mask's [Behaviour\(\)](#) method can override the background color (default = None).

`decoration` [Terminal.TextDecoration](#)

The default decoration of the mask (a mask's [Behaviour\(\)](#) method can override the decoration (default = Default).

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

This [Mask](#)'s [Behaviour\(\)](#) is to display large numbers .

```
protected override void Behaviour()
```

Class CustomMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

This [Mask](#) helps to create a [Mask](#) with unique behaviour without needing to create an entire new class.

```
public class CustomMask : Mask
```

Inheritance

[object](#)  ← [Mask](#) ← CustomMask

Inherited Members

[Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) ,
[Mask.Decoration](#) , [Mask.Draw\(\)](#) ,
[Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

CustomMask(Component, Action<CustomMask>, bool,
TextColor, BackgroundColor, TextDecoration)

Create a [Mask](#) with unique [Behaviour\(\)](#) ([action](#)).

```
public CustomMask(Component component, Action<CustomMask> action, bool isVisible  
= true, Terminal.TextColor color = TextColor.White, Terminal.BackgroundColor  
background = BackgroundColor.None, Terminal.TextDecoration decoration  
= TextDecoration.Default)
```

Parameters

component [Component](#)

The component which the mask is attached to.

action [Action](#) <[CustomMask](#)>

The [Action](#) to perform when [Behaviour\(\)](#) is called.

isVisible [bool](#)

Represent the visibility of the mask (default = true).

color [Terminal.TextColor](#)

The default color of the mask (a mask's [Behaviour\(\)](#) method can override the color (default = white).

background [Terminal.BackgroundColor](#)

The default background color of the mask (a mask's [Behaviour\(\)](#) method can override the background color (default = None).

decoration [Terminal.TextDecoration](#)

The default decoration of the mask (a mask's [Behaviour\(\)](#) method can override the decoration (default = Default).

Fields

Properties

Holds properties used in the `TUIS.Components.Masks.CustomMask._action`. (Useless for now).

```
public readonly Dictionary<string, object> Properties
```

Field Value

[Dictionary](#) <[string](#), [object](#)>

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

This [Mask](#)'s [Behaviour\(\)](#) is to call the TUIS.Components.Masks.CustomMask._action.

```
protected override void Behaviour()
```

Class ImageMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

This Mask will draw an image in format jpg, png or any other image format in ASCII art in black and white or using 8 different colors.

```
public class ImageMask : Mask
```

Inheritance

[object](#)  ← [Mask](#) ← ImageMask

Inherited Members

[Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) ,
[Mask.Decoration](#) , [Mask.Draw\(\)](#) ,
[Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

ImageMask(Component, string, ImageCharacterisation, bool, TextColor, BackgroundColor, TextDecoration)

Draw an image in format jpg, png or any other image format in ASCII art. Warning: the original image should have a bigger height and width than the [Component](#).

```
public ImageMask(Component component, string path, ImageMask.ImageCharacterisation  
characterisation = ImageCharacterisation.Brightness, bool isVisible = true,  
Terminal.TextColor color = TextColor.White, Terminal.BackgroundColor background =  
BackgroundColor.None, Terminal.TextDecoration decoration = TextDecoration.Default)
```

Parameters

component [Component](#)

The component which the mask is attached to.

path [string](#)[↗]

The relative or absolute path of the image to draw is ASCII art

characterisation [ImageMask.ImageCharacterisation](#)

Set what the different characters of the image should represent (default = [Brightness](#)).

isVisible [bool](#)[↗]

Represent the visibility of the mask (default = true).

color [Terminal.TextColor](#)

The default color of the mask (a mask's [Behaviour\(\)](#) method can override the color (default = white).

background [Terminal.BackgroundColor](#)

The default background color of the mask (a mask's [Behaviour\(\)](#) method can override the background color (default = None).

decoration [Terminal.TextDecoration](#)

The default decoration of the mask (a mask's [Behaviour\(\)](#) method can override the decoration (default = Default).

Properties

Characterisation

Holds whether the different characters of the image should represent the [Brightness](#) or the [Alpha](#) of the pixels. Trigger a [NeedReDraw](#) on value change.

```
public ImageMask.ImageCharacterisation Characterisation { get; set; }
```

Property Value

[ImageMask.ImageCharacterisation](#)

NbOfBitByColor

Holds the number of bit

```
public byte NbOfBitByColor { get; set; }
```

Property Value

[byte](#)

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

This [Mask](#)'s [Behaviour\(\)](#) is to draw the TUIS.Components.Masks.ImageMask._image in ascii art.

```
protected override void Behaviour()
```

Export(string, int?, int?)

```
public void Export(string path = "./exported_image.png", int? exportedWidth = null, int? exportedHeight = null)
```

Parameters

path [string](#)

exportedWidth [int](#)?

exportedHeight [int](#)?

Enum ImageMask.ImageCharacterisation

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

```
public enum ImageMask.ImageCharacterisation
```

Fields

Alpha = 1

Brightness = 0

Class InputMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

WIP. This mask let you use [ReadLine\(\)](#) to get some user input as command or text

```
public class InputMask : Mask
```

Inheritance

[object](#) ← [Mask](#) ← InputMask

Inherited Members

[Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) ,
[Mask.Decoration](#) , [Mask.Draw\(\)](#) ,
[Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#)

Constructors

InputMask(Component, Action<InputMask>?, byte, byte, bool,
TextColor, BackgroundColor, TextDecoration)

```
public InputMask(Component component, Action<InputMask>? onOutputChange = null, byte  
horizontalPadding = 0, byte verticalPadding = 0, bool isVisible = true,  
Terminal.TextColor color = TextColor.White, Terminal.BackgroundColor background =  
BackgroundColor.None, Terminal.TextDecoration decoration = TextDecoration.Default)
```

Parameters

component [Component](#)

onOutputChange [Action](#) <[InputMask](#)>

horizontalPadding [byte](#)

verticalPadding [byte](#)

isVisible [bool](#)

color [Terminal.TextColor](#)

background [Terminal.BackgroundColor](#)

decoration [Terminal.TextDecoration](#)

Properties

Enabled

Holds whether or not the [Mask](#) should read the user inputs.

```
public bool Enabled { get; set; }
```

Property Value

[bool](#)

HorizontalPadding

```
public byte HorizontalPadding { get; set; }
```

Property Value

[byte](#)

Output

Holds the [string](#) corresponding to the last input of the user

```
public string Output { get; }
```

Property Value

[string](#)

VerticalPadding

```
public byte VerticalPadding { get; set; }
```

Property Value

[byte](#)

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

```
protected override void Behaviour()
```

Class Mask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

A Mask must be attached to a [Component](#) and adds a layer of drawings on top of it. Masks will be drawn in the same order as they have been added to the [Masks](#) list.

```
public abstract class Mask
```

Inheritance

[object](#)  ← Mask

Derived

[BackgroundMask](#), [BoxMask](#), [ClockMask](#), [CustomMask](#), [ImageMask](#), [InputMask](#), [TextMask](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

Mask(Component, bool, TextColor, BackgroundColor, TextDecoration)

A Mask goes in a [Component](#) and will add characters to print on it following custom rules: [Behaviour\(\)](#).

```
public Mask(Component component, bool isVisible = true, Terminal.TextColor color =  
    TextColor.White, Terminal.BackgroundColor background = BackgroundColor.None,  
    Terminal.TextDecoration decoration = TextDecoration.Default)
```

Parameters

component [Component](#)

The component which the mask is attached to.

isVisible [bool](#) 

Represent the visibility of the mask (default = true).

`color` [Terminal.TextColor](#)

The default color of the mask (a mask's [Behaviour\(\)](#) method can override the color (default = white).

`background` [Terminal.BackgroundColor](#)

The default background color of the mask (a mask's [Behaviour\(\)](#) method can override the background color (default = None).

`decoration` [Terminal.TextDecoration](#)

The default decoration of the mask (a mask's [Behaviour\(\)](#) method can override the decoration (default = Default).

Fields

Component

The [Component](#) the [Mask](#) is attached to.

```
public readonly Component Component
```

Field Value

[Component](#)

Properties

Background

Keeps the current [Terminal.BackgroundColor](#) of the [Mask](#).

```
public Terminal.BackgroundColor Background { get; set; }
```

Property Value

[Terminal.BackgroundColor](#)

Color

Keeps the current [Terminal.TextColor](#) of the [Mask](#).

```
public Terminal.TextColor Color { get; set; }
```

Property Value

[Terminal.TextColor](#)

Decoration

Keeps the current [Terminal.TextDecoration](#) of the [Mask](#).

```
public Terminal.TextDecoration Decoration { get; set; }
```

Property Value

[Terminal.TextDecoration](#)

IsVisible

Represent the visibility of the [Mask](#).

```
public bool IsVisible { get; set; }
```

Property Value

[bool](#)

NeedReDraw

Helps to optimize the drawing of components, if NeedReDraw == false, the [Mask](#) won't be re-[Draw\(\)](#)n. if NeedReDraw is set to true, it will automatically set the [NeedReDraw](#) to true.

```
public bool NeedReDraw { get; set; }
```

Property Value

[bool](#)

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [DrawChar\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

```
protected abstract void Behaviour()
```

Draw()

This method will be called if the mask has changed and is visible. It calls the [Behaviour\(\)](#) method of the mask.

```
public void Draw()
```

DrawChar(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?)

Used in the [Behaviour\(\)](#) method of [Mask](#)s to properly print a character using relative coordinates.

```
protected void DrawChar(int y, int x, char? c, Terminal.TextColor? textColor = null,
Terminal.BackgroundColor? backgroundColor = null, Terminal.TextDecoration?
textDecoration = null)
```

Parameters

y [int](#)

Coordinate on the y-axis (0 = top) relative to the [Component](#).

x [int](#)

Coordinate on the x-axis (0 = left) relative to the [Component](#).

c [char](#)?

The character to print (null = no print).

textColor [Terminal.TextColor](#)?

The color which the **c** will be printed. null = the default background color of the mask.

backgroundColor [Terminal.BackgroundColor](#)?

The background color which the **c** will be printed. null = the default background color of the mask.

textDecoration [Terminal.TextDecoration](#)?

The decoration with which the **c** will be printed. null = the default decoration of the mask.

DrawChar(int, int, char?, (byte r, byte g, byte b), BackgroundColor?, TextDecoration?)

```
protected void DrawChar(int y, int x, char? c, (byte r, byte g, byte b) textColor,  
Terminal.BackgroundColor? backgroundColor = null, Terminal.TextDecoration?  
textDecoration = null)
```

Parameters

y [int](#)

x [int](#)

c [char](#)?

textColor ([byte](#) **r**, [byte](#) **g**, [byte](#) **b**)

backgroundColor [Terminal.BackgroundColor](#)?

textDecoration [Terminal.TextDecoration](#)?

Class TextMask

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

This [Mask](#) will draw literal custom text.

```
public class TextMask : Mask
```

Inheritance

[object](#)  ← [Mask](#) ← TextMask

Derived

[BigTextMask](#)

Inherited Members

[Mask.Component](#) , [Mask.NeedReDraw](#) , [Mask.IsVisible](#) , [Mask.Color](#) , [Mask.Background](#) ,
[Mask.Decoration](#) , [Mask.Draw\(\)](#) ,
[Mask.DrawChar\(int, int, char?, Terminal.TextColor?, Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[Mask.DrawChar\(int, int, char?, \(byte r, byte g, byte b\), Terminal.BackgroundColor?, Terminal.Text
Decoration?\)](#) ,
[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

TextMask(Component, string, byte, byte,
HorizontalAlignmentEnum, VerticalAlignmentEnum, bool,
TextColor, BackgroundColor, TextDecoration)

This [Mask](#) will draw some text.

```
public TextMask(Component component, string text, byte horizontalPadding = 0, byte  
verticalPadding = 0, TextMask.HorizontalAlignmentEnum horizontalAlignment =  
HorizontalAlignmentEnum.Left, TextMask.VerticalAlignmentEnum verticalAlignment =  
VerticalAlignmentEnum.Top, bool isVisible = true, Terminal.TextColor color =
```

```
TextColor.White, Terminal.BackgroundColor background = BackgroundColor.None,  
Terminal.TextDecoration decoration = TextDecoration.Default)
```

Parameters

component [Component](#)

The component which the mask is attached to.

text [string](#)

the string of text to draw.

horizontalPadding [byte](#)

the distance between the left/right sides of the **component** and the beginning/end of the **text**'s rows.

verticalPadding [byte](#)

the distance between the top/bottom sides of the **component** and the first/last row of the **text**.

horizontalAlignment [TextMask.HorizontalAlignmentEnum](#)

Whether the **text** should be aligned to the [Left](#), [Center](#) or [Right](#) of the **component**.

verticalAlignment [TextMask.VerticalAlignmentEnum](#)

Whether the **text** should be aligned to the [Top](#), [Center](#) or [Bottom](#) of the **component**.

isVisible [bool](#)

Represent the visibility of the mask (default = true).

color [Terminal.TextColor](#)

The default color of the mask (a mask's [Behaviour\(\)](#) method can override the color (default = white).

background [Terminal.BackgroundColor](#)

The default background color of the mask (a mask's [Behaviour\(\)](#) method can override the background color (default = None).

decoration [Terminal.TextDecoration](#)

The default decoration of the mask (a mask's [Behaviour\(\)](#) method can override the decoration (default = Default).

Properties

HorizontalAlignment

Holds the current [TextMask.HorizontalAlignmentEnum](#) of the [Text](#).

```
public TextMask.HorizontalAlignmentEnum HorizontalAlignment { get; set; }
```

Property Value

[TextMask.HorizontalAlignmentEnum](#)

HorizontalPadding

Holds the distance between the left/right sides of the [Component](#) and the beginning/end of the [Text](#)'s rows.

```
public byte HorizontalPadding { get; set; }
```

Property Value

[byte](#) 

Text

Holds the string of text to draw.

```
public string Text { get; set; }
```

Property Value

[string](#) 

VerticalAlignment

Holds the current [TextMask.VerticalAlignmentEnum](#) of the [Text](#);

```
public TextMask.VerticalAlignmentEnum VerticalAlignment { get; set; }
```

Property Value

[TextMask.VerticalAlignmentEnum](#)

VerticalPadding

Holds the distance between the top/bottom sides of the [Component](#) and the first/last row of the [Text](#).

```
public byte VerticalPadding { get; set; }
```

Property Value

[byte](#) 

Methods

Behaviour()

This method is what makes the mask unique, it will describe that to print and where using the [Draw Char\(int, int, char?, TextColor?, BackgroundColor?, TextDecoration?\)](#). The method will automatically be called when needed.

```
protected override void Behaviour()
```

Enum TextMask.HorizontalAlignmentEnum

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

```
public enum TextMask.HorizontalAlignmentEnum
```

Fields

Center = 1

Left = 0

Right = 2

Enum TextMask.VerticalAlignmentEnum

Namespace: [TUIS.Components.Masks](#)

Assembly: TUIS.dll

```
public enum TextMask.VerticalAlignmentEnum
```

Fields

Bottom = 2

Center = 1

Top = 0

Namespace TUIS.Systems

Classes

[InputSystem](#)

[TimeSystem](#)

Class InputSystem

Namespace: [TUIS.Systems](#)

Assembly: TUIS.dll

```
public class InputSystem : IDisposable
```

Inheritance

[object](#)  ← InputSystem

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

Start()

```
public void Start()
```

Events

OnKeyPress

```
public event Action<ConsoleKeyInfo>? OnKeyPress
```

Event Type

[Action](#) <[ConsoleKeyInfo](#)>

Class TimeSystem

Namespace: [TUIS.Systems](#)

Assembly: TUIS.dll

```
public class TimeSystem
```

Inheritance

[object](#)  ← TimeSystem

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

TimeSystem()

```
public TimeSystem()
```

Fields

Timer

```
public readonly Timer Timer
```

Field Value

[Timer](#) 

Properties

MiliSecond

```
public byte MiliSecond { get; }
```

Property Value

[byte](#)

Methods

AddTimedEvent(ElapsedEventHandler?)

```
public void AddTimedEvent(ElapsedEventHandler? eventHandler)
```

Parameters

eventHandler [ElapsedEventHandler](#)

Start()

```
public void Start()
```